Rockwell Automation Allen-Bradley PowerFlex 525 User Manual

Product: PowerFlex 525 Adjustable Frequency AC Drive - Frame A.

Power Rating: 1.5 kW (2.0 HP) output at 3-Phase 380-480V AC input, 4.0A continuous output.

Application: Variable speed regulation for CONV01 infeed belt conveyor system.

Input Power: Supplied from circuit breaker Q3 on schematic E205 Cell Power Distribution.

Control Wiring: Terminal 01 (Stop), Terminal 02 (Start), Terminal 04 (Fault Reset from PLC01).

Analog Speed Reference: Terminal 13 (0-10V DC analog input), Terminal 14 (Analog common).

PowerFlex 525 Drive Fault Diagnostics - Fault F004 Overcurrent

Trip Description: The drive output current has exceeded the hardware instantaneous current limit.

Possible Cause 1: Motor winding short circuit or phase-to-ground insulation breakdown on CONV01 motor.

Possible Cause 2: Excessive mechanical load on CONV01 caused by seized roller bearings or jammed forgings

Possible Cause 3: Acceleration time parameter P041 Accel Time 1 set too short (< 2.5 seconds).

Diagnostic Action: Perform Megger insulation resistance test at 1000V DC. Resistance must exceed 50 MOhm.

Verify free mechanical rotation of conveyor drive roller before restarting production.

PowerFlex 525 Speed Reference and Analog Calibration

Parameter P047 [Speed Reference 1] must be programmed to Option 1 (0-10V Input).

Parameter P043 [Minimum Frequency] = 0.0 Hz; Parameter P044 [Maximum Frequency] = 50.0 Hz.

Speed drift under load on CONV01 can result from analog reference voltage droop across long cabling runs.

Ensure foil shield of analog signal cable is grounded at cabinet terminal strip per schematic E205.

Dynamic braking resistor connected across terminals BR+ and BR- for rapid conveyor deceleration.